

Lecture 2:

Mathematical Analysis

1 Intervals

Definition 1.1 (Interval). [def:interval] A set $I \subseteq \mathbb{R}$ is an *interval* if, whenever $x, y \in I$ with $x \leq y$, every $z \in \mathbb{R}$ satisfying $x \leq z \leq y$ also belongs to I .

Remark 1.1 (Intervals as Sets). [rem:intervals-as-sets] In these notes, every interval is a set contained in $\mathbb{R}$; that is, every interval I satisfies $I \subseteq \mathbb{R}$.

Definition 1.2 (Open Interval). [def:open-interval] For $a, b \in \mathbb{R}$ with $a < b$, the *open interval* from a to b is

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}.$$

Definition 1.3 (Closed Interval). [def:closed-interval] For $a, b \in \mathbb{R}$ with $a \leq b$, the *closed interval* from a to b is

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}.$$

2 Nested Intervals Property

Definition 2.1 (Nested Intervals Property (NIP)). [def:nip] Let $I_n = [a_n, b_n]$ be a sequence of closed and bounded intervals.